



Fig. 4: Configuration bit for the microcontroller

is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB as it saves time and minimises assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC base for the microcontroller.

Before inserting IC1, check the supply voltage at test point TP1. It should be 5V. A reflector as available for high power LED will direct the flash properly to the object under examination.

Cut a round disk of cardboard and make a dark patch on it. Fit the disk onto the motor spindle. When you power-on the motor, the disk starts rotating. Now power-on the circuit. The high-power LED starts flashing. Direct flashing light from the stroboscope onto the rotating marked disk. Adjust the LED flashing rate by varying trimpot VR1 until the patch appears stationary. Flashing light's pulse time period in milliseconds (ms) is displayed on the 7-segment display. You can see the pulse rate on the display by pressing switch S5.

To test the circuit for proper functioning, check test points for correct voltage levels as shown in the test point table.

Software

The program is written in Assembly language and assembled using PIC simulator IDE of Oshon software project. It is well commented and easy to understand. The generated hex code is burnt into the microcontroller using a suitable programmer. The simulator provides all the accessories for simulation of the program, such as microcontroller pins, seven-segment display

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

and also a square wave. The PIC Kit 2 programmer display is shown in Fig. 4 with chosen configuration bits.

The program for the stroboscope with these functions and a display is fairly complex. The timer of the PIC is employed to generate an interrupt once every 256 clock cycles of the timer. The timer clock is derived from the system clock by a pre-scaler, which is contained in 'option register' of the chip.

Port C is fully utilised for pulse generation, mode selection and display selection. Port B is used for driving the seven segments and the decimal point. One ADC channel is used here. Using a multi-turn potmeter, the voltage between 0 and 5V is applied to analogue input—pin RA0 of IC1. This serves to give a smooth input for adjusting the pulse rate of flashing LED2.

The flash pulse comes from port pin RC0. This pulse width is to be adjusted in proportion to the rate. Too much width will cause smearing while freezing a moving object. The pulse rate is varied by adjusting the number of times the timer interrupt should elapse before starting a new pulse. Thus the number is variable from 0 to 255.

After initialising the timer and ports, the program checks the status of port pin RC6. Depending on this status, the program decides whether a variable pulse or a fixed 50Hz pulse is to be used. For a fixed 50Hz pulse, a 50Hz square wave of 5 V is applied to port pin RC5.

Display segments are selected one by one. The program uses a lookup table to define the various segment patterns to be shown (0 to 9). Selection of the digits is done through port pins RC1 through RC3. ●

Prof. K. Padmanabhan is an emeritus professor at Alagappa College of Technology, Guindy, Chennai, and Dr S. Ananthi is head of instrumentation at the Madras University

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